Assessing Unintended Memorization in Language Models through Semantic Leakage

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Abstract

Machine Learning models can unintentionally memorize sensitive information during training, which poses huge risks even when critical data is not explicitly stated. Designing safer systems, especially in domains such as healthcare or finance, requires a good understanding and evaluation of how models retain both direct and indirect information. In this work, we aim to evaluate these risks by distinguishing between literal memorization and semantic leakage by quantifying unintended memorization in neural language models using the exposure metric. We aim to investigate whether high exposure scores or memorization stem from the semantic priors in pre-trained word embeddings rather than genuine memorization of the training data. Our results distinguish between genuine memorization and embedding-driven generalization, demonstrating that exposure can be a confounded metric for privacy auditing and emphasizing the need for more nuanced evaluation frameworks.

1 Introduction

The extent to which modern neural language models memorize training data is a critical concern for privacy-preserving deployment. Previous work [1] utilized the "exposure" metric to quantify unintended memorization in language models, showing that rare sequences inserted during training can be extracted with high probability. While this metric is crucial, it might not distinguish between genuine memorization and the effects of semantic priors encoded in pre-trained embeddings. In this work, we interrogate the validity of exposure as a standalone metric and we hypothesize that high exposure scores may not solely reflect memorization but can also arise from semantic generalization derived from pre-trained word embeddings. If embeddings encode strong semantic priors, a model may correctly predict a "secret" based on geometric proximity in the embedding space rather than actual exposure to the data. For example, a model may infer a "secret" disease statement with a non-zero probability even without encountering the secret information during training.

To test our hypothesis[1], we replicate the work of [1] using recurrent neural networks trained with and without differentially private SGD, by varying the embedding dimensions and controlling for semantic similarity. By comparing exposure scores across these various conditions, we aim to reliably evaluate exposure as a privacy auditing metric and motivate post-training interventions to mitigate privacy risks in sensitive applications.

2 Background and Related Work

Privacy risks in machine learning are shifting from concerns over exact data reconstruction to "semantic privacy", where the threat lies in a model's ability to

[1]For our project, we selected the paper reading option by studying related papers, sharing knowledge, and exploring a new hypothesis.

infer sensitive concepts rather than just regurgitate them verbatim. Traditional research has focused on the tendency of Large Language Models (LLMs) to memorize training data exactly. Carlini et al. (2023) quantified this phenomenon, establishing log-linear relationships between model scale, data duplication, and context length [2]. Standard Membership Inference Attacks (MIAs) rely on this "memorization effect", using overfitting signals to distinguish members from non-members [3]. However, these metrics struggle when models undergo distillation, as demonstrated in [3] where it was shown that while instance-level Membership Inference Attacks (MIAs) fail against distilled models, distributional statistics reveal a persisting "memory chain" of sensitive information.

Recent work highlights semantic attacks as a distinct threat model bridging the gap left by exact reconstruction studies. The authors in [4] define "semantic privacy" as protecting attributes that are inferred through contextual reasoning rather than explicitly stated. This creates vulnerabilities even when Personal Identifiable Information (PII) is removed. [5] introduced the Semantic Membership Inference Attack (SMIA), showing that models exhibit distinguishable probability distributions on semantically similar inputs (neighbors) even without emitting the exact training string. The definition of memorization has been evolving to account for these factors. [6] propose a "multi-prefix" framework, arguing that deeply memorized sequences are characterized by the diversity of prefixes that elicit them, rather than just single-path retrieval. Furthermore, mitigation strategies like "goldfish loss" successfully prevent verbatim reproduction but fail to stop the generation of semantic paraphrases [7].

Building on previous work in [1], we investigate how pre-trained word embeddings might contribute to high exposure scores, which may reflect semantic priors instead of genuine memorization. [1] investigate how the source and dimension of pretrained word embeddings influence memorization in neural language models. To quantify this, the authors utilize the "exposure" metric introduced by [8], which measures a secret's extractability by comparing its log-likelihood against a set of plausible candidates. They studied training a single-layer LSTM using GloVe, ELMo, and BERT embeddings on the Penn TreeBank dataset augmented with synthetic secrets (disease statements and PINs) inserted at varying frequencies. The authors report three primary findings: (i) larger embedding dimensions correlate with higher exposure values, indicating a stronger capacity to memorize rare sequences; (ii) inserting multiple distinct secrets of the same category reduces the exposure of any single instance, as probability mass is distributed across similar sequences; and (iii) differentially private stochastic gradient descent (DP-SGD) significantly mitigates exposure, providing a robust defense against unintended memorization.

3 Problem Statement and Setup

While using the exposure metric to quantify unintended memorization is standard, it may conflate two distinct phenomena: (i) Literal memorization, where the model reproduces exact sentences seen during training and (ii) Semantic leakage, where the model predicts sensitive information using semantic context embedded in pre-trained embeddings.

To disentangle these factors, we build on [1] by designing experiments with the following controlled variables: (a) **Secret Type:** we evaluate *Disease statements* (e.g., "John is suffering from Alzheimer") and *PINs* (e.g., "John ATM pin is 1234"). (b) **Insertion Mode:** we compare *Direct insertion* (explicit secret in training data) against *Semantic insertion* (e.g., "John forgets names frequently"). In the semantic case, the sensitive token ("Alzheimer") never appears in the training set, allowing us to isolate embedding-driven inference. (c) **Secret Density:** secrets were inserted either individually or in clusters to identify interference effects. (d) **Embedding & Privacy:** we investigate static (GloVe) vs. contextual (BERT) embeddings, trained with and without DP-SGD. Owing to limited computational resources, training duration was calibrated: 5 epochs for disease secrets (12 with DP) and 40 epochs for PINs (100 with DP). We compute exposure using the rank-based formula derived by [8]. By extending this analysis to include semantic insertion modes, we can evaluate whether high exposure scores indicate genuine data leakage or merely the retrieval of public semantic priors.

4 Results and Discussion

We present our experimental results in Table 1 and Fig. 1. Our analysis distinguishes between literal memorization and semantic leakage across varying embedding types and privacy settings. Consistent with [1], **Direct Single Insertions** yield the highest exposure scores (e.g., 13.29 for PINs), indicating strong literal memorization. Contrary to [1], our results across models (see Figure 1) suggest this memorization is largely invariant under embedding dimension, implying that the capacity to overfit specific secrets saturates even at lower dimensions. We also confirm the "interference" hypothesis. As shown in Fig. 1, moving from **single** to **multiple** insertions consistently reduces exposure in the non-DP setting. For PINs, exposure drops from 13.29 to 10.97, confirming that probability mass is distributed across competing candidates, acting as a regularizer.

The results also reveal our most critical finding: **semantic insertions** exhibit non-zero exposure scores (e.g., ~ 2.20 for PINs and ~ 1.98 for Disease) even when the secret was never explicitly seen during training. This confirms that the exposure metric captures **semantic leakage**. High exposure scores in privacy audits may thus be false positives confounded by semantic priors. We stress-tested DP-SGD on the BERT-1024d model (Table 1). While DP successfully reduced PIN exposure to near-zero (~ 0.007), validating its effectiveness against literal memorization, it failed to mitigate exposure for Disease secrets. Surprisingly, Disease exposure did not decrease (~ 4.05) but instead increased under DP with a privacy budget ($\epsilon \approx 32$), unlike in other experiments. This divergence suggests that while DP-SGD obscures exact training paths, it might struggle to suppress robust semantic priors (e.g., symptom-disease associations) embedded in the pre-trained model; this phenomenon requires further investigation in future work.

Secret	Insert	Mode	Exposure (No DP)	Exposure (DP)
Disease	Multiple	Direct	4.7901	4.0531
Disease	Multiple	Semantic	2.0531	4.0531
Disease	Single	Direct	4.7901	4.0531
Disease	Single	Semantic	1.9827	4.0531
PIN	Multiple	Direct	10.9658	0.0074
PIN	Multiple	Semantic	2.2304	0.0016
PIN	Single	Direct	13.2877	0.0077
PIN	Single	Semantic	2.2036	0.0016

Table 1: Exposure results for BERT-1024d with and without DP.

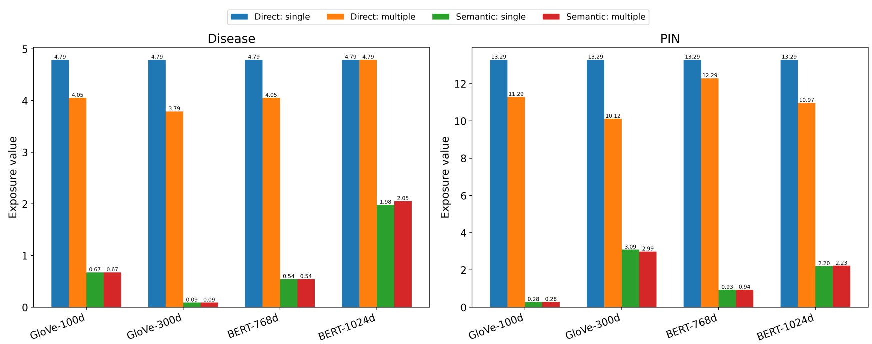

Figure 1: Exposure scores for secrets under various scenarios.

5 Conclusion and Future Work

In this work, we demonstrated that the "exposure" metric is a confounded measure for privacy auditing, as it fails to reliably distinguish between genuine memorization of training data and "semantic leakage" derived from pre-trained embeddings. Our experiments revealed that models exhibit non-zero exposure scores for semantically similar secrets they never encountered during training, indicating that high exposure often reflects the retrieval of public semantic priors rather than exact memorization. Furthermore, while DP successfully mitigates literal memorization (e.g., PINs) in our experiments, it proves ineffective against semantic leakage, leaving robust prior associations (e.g., disease symptoms) exposed. Consequently, we conclude that current privacy frameworks must evolve to disentangle these two phenomena to prevent false positives in privacy audits. Future work includes exploring the phenomena observed in this study, and developing post-training interventions in LLMs that modify embedding-space representations to steer sensitive data towards semantically safe embedding that reduces literal and semantic leakage.

References

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6 Appendix

Dataset All experiments use the Penn TreeBank (PTB) dataset with predefined train, validation, and test splits. Text is lower-cased and tokenized using a custom tokenizer and the training data is segmented into regular sequences with a length of 35 tokens. The vocabulary is built from the PTB tokens, many disease names, all possible 4-digit PINs, and all tokens appearing in secret sentences. A maximum number of sentences can be enforced to control dataset size (2000 sentences as performed in [1]). Secrets are inserted either directly where the secret is explicitly stated or semantically using natural language that imply the secret without explicitly stating it. Each canary is inserted exactly once in the whole dataset.

Model Architecture and Training Configuration We use a single layer LSTM for training. This model also consists of an embedding layer with an initialization from pre-trained embeddings (GloVe or BERT). Adam optimizer is used with a learning rate of 1×10^{-3}, batch size of 64. The objective is token level cross-entropy loss. For DP-experiments, we use an Opacus-based framework where gradients are clipped and perturbed with a Gaussian noise. We set $\delta = 2 \times 10^{-5}$.

Evaluation Privacy leakage is quantified using the exposure metric. Given a context prefix, the model scores all candidates, and exposure is defined as

$$\text{exposure} = \log_2 |\mathcal{R}| - \log_2 r$$

Where r is the true secret rank among a candidate set $\mathcal{R}$. High exposure values indicate strong memorization of the secret by the model.